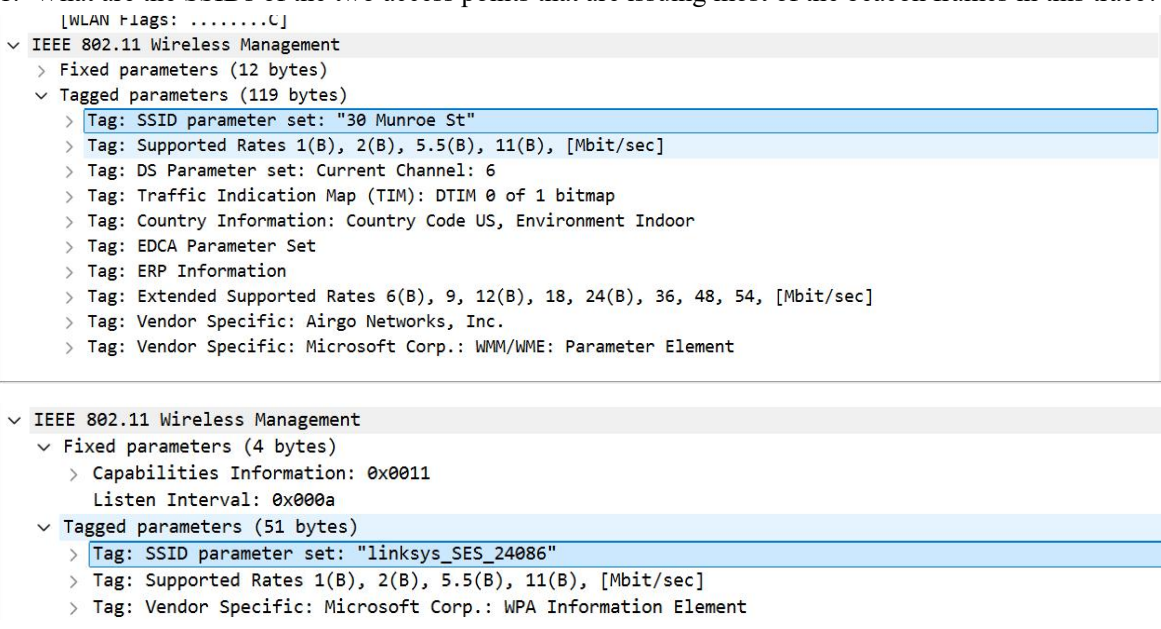
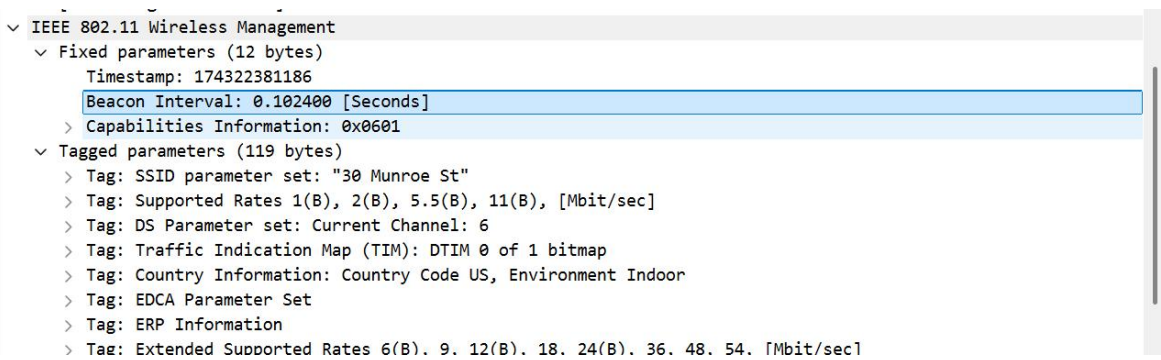


山东大学 计算机科学与技术 学院

新兴网络技术与实践 课程实验报告

学号：202300130183	姓名：宋浩宇	班级：23 级智能班
实验题目：Wireshark Lab: 802.11 WiFi v8.0		
实验学时：4	实验日期：2025/5/28	
实验目的：学习了解 WiFi		
实验结果： 1. What are the SSIDs of the two access points that are issuing most of the beacon frames in this trace?  如图，是”39 Munroe St”和”linksys_SES_24086” 2. What are the intervals of time between the transmissions of the beacon frames the linksys_ses_24086 access point? From the 30 Munroe St. access point? (Hint: this interval of time is contained in the beacon frame itself). 		

```

IEEE 802.11 Wireless Management
  Fixed parameters (12 bytes)
    Timestamp: 9534922650285
    Beacon Interval: 0.102400 [Seconds]
  Capabilities Information: 0x0011
  Tagged parameters (26 bytes)
    Tag: SSID parameter set: "linksys12"
    Tag: Supported Rates 1(B), 2(B), 5.5, 11, [Mbit/sec]
    Tag: DS Parameter set: Current Channel: 6
    Tag: Traffic Indication Map (TIM): DTIM 1 of 3 bitmap

```

如图，两个的时间间隔都是 0.102400s

3. What (in hexadecimal notation) is the source MAC address on the beacon frame from 30 Munroe St? Recall from Figure 7.13 in the text that the source, destination, and BSS are three addresses used in an 802.11 frame. For a detailed discussion of the 802.11 frame structure, see section 7 in the IEEE 802.11 standards document (cited above).

```

> Frame 1: 183 bytes on wire (1464 bits), 183 bytes captured (1464 bits)
> Radiotap Header v0, Length 24
> 802.11 radio information
IEEE 802.11 Beacon frame, Flags: .....C
  Type/Subtype: Beacon frame (0x0008)
  Frame Control Field: 0x8000
  .000 0000 0000 0000 = Duration: 0 microseconds
  Receiver address: Broadcast (ff:ff:ff:ff:ff:ff)
  Destination address: Broadcast (ff:ff:ff:ff:ff:ff)
  Transmitter address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  Source address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  BSS Id: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  .... .... 0000 = Fragment number: 0
  1011 0010 0110 .... = Sequence number: 2854
  Frame check sequence: 0x057e2608 [unverified]
  [FCS Status: Unverified]
  [WLAN Flags: .....C]

```

如图，源 MAC 地址是 00:16:b6:f7:1d:51

4. What (in hexadecimal notation) is the destination MAC address on the beacon frame from 30 Munroe St??

```

> Frame 1: 183 bytes on wire (1464 bits), 183 bytes captured (1464 bits)
> Radiotap Header v0, Length 24
> 802.11 radio information
IEEE 802.11 Beacon frame, Flags: .....C
  Type/Subtype: Beacon frame (0x0008)
  Frame Control Field: 0x8000
  .000 0000 0000 0000 = Duration: 0 microseconds
  Receiver address: Broadcast (ff:ff:ff:ff:ff:ff)
  Destination address: Broadcast (ff:ff:ff:ff:ff:ff)
  Transmitter address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  Source address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  BSS Id: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  .... .... 0000 = Fragment number: 0
  1011 0010 0110 .... = Sequence number: 2854
  Frame check sequence: 0x057e2608 [unverified]
  [FCS Status: Unverified]
  [WLAN Flags: .....C]
IEEE 802.11 Wireless Management
  Fixed parameters (12 bytes)
    Timestamp: 9534922650285
    Beacon Interval: 0.102400 [Seconds]
  Capabilities Information: 0x0011
  Tagged parameters (26 bytes)
    Tag: SSID parameter set: "linksys12"
    Tag: Supported Rates 1(B), 2(B), 5.5, 11, [Mbit/sec]
    Tag: DS Parameter set: Current Channel: 6
    Tag: Traffic Indication Map (TIM): DTIM 1 of 3 bitmap

```

如图，目标 MAC 地址是 ff:ff:ff:ff:ff:ff

5. What (in hexadecimal notation) is the MAC BSS id on the beacon frame from 30 Munroe St?

```

> 802.11 radio information
v IEEE 802.11 Beacon frame, Flags: .....C
  Type/Subtype: Beacon frame (0x0008)
  > Frame Control Field: 0x8000
    .000 0000 0000 0000 = Duration: 0 microseconds
  > Receiver address: Broadcast (ff:ff:ff:ff:ff:ff)
  > Destination address: Broadcast (ff:ff:ff:ff:ff:ff)
  > Transmitter address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  > Source address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  > BSS Id: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
    .... 0000 = Fragment number: 0
    1011 0010 0110 .... = Sequence number: 2854
    Frame check sequence: 0x057e2608 [unverified]
    [FCS Status: Unverified]
    [WLAN Flags: .....C]
v IEEE 802.11 Wireless Management
  v Fixed parameters (12 bytes)

```

如图，BSS 的地址是 00:16:b6:f7:1d:51

6. The beacon frames from the 30 Munroe St access point advertise that the access point can support four data rates and eight additional “extended supported rates.” What are these rates?

```

[WLAN Flags: .....C]
v IEEE 802.11 Wireless Management
  v Fixed parameters (12 bytes)
    Timestamp: 174319001986
    Beacon Interval: 0.102400 [Seconds]
    > Capabilities Information: 0x0601
  v Tagged parameters (119 bytes)
    > Tag: SSID parameter set: "30 Munroe St"
    > Tag: Supported Rates 1(B), 2(B), 5.5(B), 11(B), [Mbit/sec]
    > Tag: DS Parameter set: Current Channel: 6
    > Tag: Traffic Indication Map (TIM): DTIM 0 of 1 bitmap
    > Tag: Country Information: Country Code US, Environment Indoor
    > Tag: EDCA Parameter Set
    > Tag: ERP Information
    > Tag: Extended Supported Rates 6(B), 9, 12(B), 18, 24(B), 36, 48, 54, [Mbit/sec]
    > Tag: Vendor Specific: Airgo Networks, Inc.
    > Tag: Vendor Specific: Microsoft Corp.: WMM/WME: Parameter Element

```

如图，这些速度分别是 1Mbps、2Mbps、5.5Mbps、11Mbps、6Mbps、9Mbps、12Mbps、18Mbps、24Mbps、36Mbps、48Mbps、54Mbps

7. Find the 802.11 frame containing the SYN TCP segment for this first TCP session (that downloads alice.txt). What are three MAC address fields in the 802.11 frame? Which MAC address in this frame corresponds to the wireless host (give the hexadecimal representation of the MAC address for the host)? To the access point? To the first-hop router? What is the IP address of the wireless host sending this TCP segment? What is the destination IP address? Does this destination IP address correspond to the host, access point, first-hop router, or some other network-attached device? Explain.

```

v IEEE 802.11 QoS Data, Flags: .....TC
  Type/Subtype: QoS Data (0x0028)
  > Frame Control Field: 0x8801
    .000 0000 0010 1100 = Duration: 44 microseconds
  > Receiver address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  > Transmitter address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)
  > Destination address: CiscoLinksys_f4:eb:a8 (00:16:b6:f4:eb:a8)
  > Source address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)
  > BSS Id: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  > STA address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)
    .... 0000 = Fragment number: 0
    0000 0011 0001 .... = Sequence number: 49
    Frame check sequence: 0xad57fce0 [unverified]
    [FCS Status: Unverified]
    [WLAN Flags: .....TC]
  > Qos Control: 0x0000

```

如图，主机是 00:13:02:d1:b6:4f，接入点是 00:16:b6:f7:1d:51，第一跳地址是 00:16:b6:f4:eb:a8，


```

Internet Protocol Version 4, Src: 192.168.1.109, Dst: 128.119.245.12
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 48
  Identification: 0x1324 (4900)
  > 010. .... = Flags: 0x2, Don't fragment
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 128
  Protocol: TCP (6)
  Header Checksum: 0xb00a [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.1.109
  Destination Address: 128.119.245.12
  [Stream index: 1]

```

如图，发送这个 TCP 段的无线主机的 ip 地址是 192.168.1.109，目的地址是 128.119.245.12，该目的 ip 对应的应该是其他联网设备

8. Find the 802.11 frame containing the SYNACK segment for this TCP session. What are three MAC address fields in the 802.11 frame? Which MAC address in this frame corresponds to the host? To the access point? To the first-hop router? Does the sender MAC address in the frame correspond to the IP address of the device that sent the TCP segment encapsulated within this datagram? (Hint: review Figure 6.19 in the text if you are unsure of how to answer this question, or the corresponding part of the previous question. It's particularly important that you understand this).

```

[Duration: 36µs]
IEEE 802.11 QoS Data, Flags: ..mP..F.C
  Type/Subtype: QoS Data (0x0028)
  > Frame Control Field: 0x8832
  Duration/ID: 11560 (reserved)
  > Receiver address: 91:2a:b0:49:b6:4f (91:2a:b0:49:b6:4f)
  > Transmitter address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  > Destination address: 91:2a:b0:49:b6:4f (91:2a:b0:49:b6:4f)
  > Source address: CiscoLinksys_f4:eb:a8 (00:16:b6:f4:eb:a8)
  > BSS Id: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  > STA address: 91:2a:b0:49:b6:4f (91:2a:b0:49:b6:4f)
  .... 0000 = Fragment number: 0
  1100 0011 0100 .... = Sequence number: 3124
  Frame check sequence: 0xecdc407d [unverified]
  [FCS Status: Unverified]
  [WLAN Flags: ..mP..F.C]
  > Qos Control: 0x0100
  > Logical-Link Control
Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.1.109
  0100 .... = Version: 4

```

如图，源地址是 00:16:b6:f4:eb:a8，接入点是 00:16:b6:f7:1d:51，第一跳地址是 91:2a:b0:49:b6:4f，发送此 TCP 段的无线主机的 ip 地址是 128.119.245.12，目的 ip 地址是 192.168.1.109，该目的 ip 地址是主机。

9. What two actions are taken (i.e., frames are sent) by the host in the trace just after t=49, to end the association with the 30 Munroe St AP that was initially in place when trace collection began? (Hint: one is an IP-layer action, and one is an 802.11-layer action). Looking at the 802.11 specification, is there another frame that you might have expected to see, but don't see here?

1732 49.542481	CiscoLinksys_f7:1d:51	Broadcast	802.11	183 Beacon frame, SN=3588, FN=0, Flags=.....C, BI=100, SSID="30 Munroe St"
1733 49.583615	192.168.1.109	192.168.1.1	DHCP	390 DHCP Release - Transaction ID 0xea5a526
1734 49.583771	Intel_d1:b6:4f	802.11	38 Acknowledgement, Flags=.....C	
1735 49.609617	Intel_d1:b6:4f	CiscoLinksys_f7:1d:51	802.11	54 Deauthentication, SN=1605, FN=0, Flags=.....C

如图，在 ip 层进行了 dhcp release

1734 49.583771	Intel_d1:b6:4f	802.11	38 Acknowledgement, Flags=.....C	
1735 49.609617	Intel_d1:b6:4f	CiscoLinksys_f7:1d:51	802.11	54 Deauthentication, SN=1605, FN=0, Flags=.....C
1736 49.609770	Intel_d1:b6:4f	802.11	38 Acknowledgement, Flags=.....C	

如图，在 802.11 层进行了 Deauthentication

10. Examine the trace file and look for AUTHENTICATION frames sent from the host to an AP and vice versa. How many AUTHENTICATION messages are sent from the wireless host to the linksys_ses_24086 AP (which has a MAC address of Cisco_Li_f5:ba:bb) starting at around t=49? .

1739	49.617713	CiscoLinksys_f5:ba:...	802.11	38 Acknowledgement, Flags=.....C
1740	49.638857	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	58 Authentication, SN=1606, FN=0, Flags=.....C
1741	49.639700	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	58 Authentication, SN=1606, FN=0, Flags=....R...C
1742	49.640702	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	58 Authentication, SN=1606, FN=0, Flags=....R...C
1743	49.641910	CiscoLinksys_f5:ba:...	802.11	38 Acknowledgement, Flags=.....C
1744	49.642315	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	58 Authentication, SN=1606, FN=0, Flags=....R...C
1745	49.644710	CiscoLinksys_f7:1d:...	Broadcast	183 Beacon frame, SN=3589, FN=0, Flags=.....C, BI=100, SSID="30 Munro
1746	49.645319	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	58 Authentication, SN=1606, FN=0, Flags=....R...C

如图，发送了 6 个 Authentication

11. Does the host want the authentication to require a key or be open?

1763	49.746105	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	EAPOL	185 Key (Message 2 of 4)
1764	49.747831	CiscoLinksys_f7:1d:...	Broadcast	802.11	183 Beacon frame, SN=3590, FN=0, Flags=.....C, BI=100, SSID="30 Munro
1765	49.749453	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	EAPOL	185 Key (Message 2 of 4)
1766	49.753595	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	EAPOL	185 Key (Message 2 of 4)
1767	49.840613	CiscoLinksys_f7:1d:...	Broadcast	802.11	183 Beacon frame, SN=3591, FN=0, Flags=.....C, BI=100, SSID="30 Munro

主机需要一个 key

12. Do you see a reply AUTHENTICATION from the linksys_ses_24086 AP in the trace?

1819	53.757881	CiscoLinksys_f5:ba:...	802.11	38 Acknowledgement, Flags=.....C	
1820	53.761198	Intel_d1:b6:4f	Broadcast	802.11	99 Probe Request, SN=1612, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1821	53.785833	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	58 Authentication, SN=1612, FN=0, Flags=.....C
1822	53.787070	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	58 Authentication, SN=1612, FN=0, Flags=....R...C
1823	53.788837	CiscoLinksys_f5:ba:...	802.11	38 Acknowledgement, Flags=.....C	

yes

13. Now let's consider what happens as the host gives up trying to associate with the linksys_ses_24086 AP and now tries to associate with the 30 Munroe St AP. Look for AUTHENTICATION frames sent from the host to and AP and vice versa. At what times are there an AUTHENTICATION frame from the host to the 30 Munroe St. AP, and when is there a reply AUTHENTICATION sent from that AP to the host in reply? (Note that you can use the filter expression "wlan.fc.subtype == 1 and wlan.fc.type == 0 and wlan.addr == IntelCor_d1:b6:4f" to display only the AUTHENTICATION frames in this trace for this wireless host.)

2155	63.161272	CiscoLinksys_f7:1d:...	Broadcast	802.11	183 Beacon frame, SN=3725, FN=0, Flags=.....C, BI=100, SSID="
2156	63.168087	Intel_d1:b6:4f	CiscoLinksys_f7:1d:...	802.11	58 Authentication, SN=1647, FN=0, Flags=.....C
2157	63.168222	Intel_d1:b6:4f	Intel_d1:b6:4f	802.11	38 Acknowledgement, Flags=.....C
2158	63.169071	Intel_d1:b6:4f	Intel_d1:b6:4f	802.11	58 Authentication, SN=3726, FN=0, Flags=.....C
2159	63.169071	CiscoLinksys_f7:1d:...	Broadcast	802.11	38 Acknowledgement, Flags=.....C

如图，在时间 63.168087 主机向接入点发出认证，在时间 63.169071 主机从接入点收到回复

14. An ASSOCIATE REQUEST from host to AP, and a corresponding ASSOCIATE RESPONSE frame from AP to host are used for the host to associated with an AP. At what time is there an ASSOCIATE REQUEST from host to the 30 Munroe StAP? When is the corresponding ASSOCIATE REPLY sent? (Note that you can use the filter expression "wlan.fc.subtype < 2 and wlan.fc.type == 0 and wlan.addr == IntelCor_d1:b6:4f" to display only the ASSOCIATE REQUEST and ASSOCIATE RESPONSE frames for this trace.)

2161	63.169910	Intel_d1:b6:4f	Intel_d1:b6:4f	802.11	80 Association Request, SN=1648, FN=0, Flags=.....C, SSID="30 Munro
2162	63.169910	Intel_d1:b6:4f	Intel_d1:b6:4f	802.11	38 Acknowledgement, Flags=.....C
2163	63.170008	Intel_d1:b6:4f	Intel_d1:b6:4f	802.11	38 Acknowledgement, Flags=.....C
2164	63.170692	CiscoLinksys_f7:1d:...	Intel_d1:b6:4f	802.11	58 Authentication, SN=3727, FN=0, Flags=.....C
2165	63.171000	CiscoLinksys_f7:1d:...	Intel_d1:b6:4f	802.11	38 Acknowledgement, Flags=.....C
2166	63.192101	CiscoLinksys_f7:1d:...	Intel_d1:b6:4f	802.11	94 Association Response, SN=3728, FN=0, Flags=.....C
2167	63.192956	CiscoLinksys_f7:1d:...	Intel_d1:b6:4f	802.11	38 Acknowledgement, Flags=.....C

如图，关联请求帧是在时间 63.169910 发出的，关联是在时间 63.192101 建立的。

15. What transmission rates is the host willing to use? The AP? To answer this question, you will need to look into the parameters fields of the 802.11 wireless LAN management frame.

```

    Tagged parameters (119 bytes)
    > Tag: SSID parameter set: "30 Munroe St"
    > Tag: Supported Rates 1(B), 2(B), 5.5(B), 11(B), [Mbit/sec]
    > Tag: DS Parameter set: Current Channel: 6
    > Tag: Traffic Indication Map (TIM): DTIM 0 of 1 bitmap
    > Tag: Country Information: Country Code US, Environment Indoor
    > Tag: EDCA Parameter Set
    > Tag: ERP Information
    > Tag: Extended Supported Rates 6(B), 9, 12(B), 18, 24(B), 36, 48, 54, [Mbit/sec]
    > Tag: Vendor Specific: Airgo Networks, Inc.
    > Tag: Vendor Specific: Microsoft Corp.: WMM/WME: Parameter Element

    IEEE 802.11 Wireless Management
    Tagged parameters (42 bytes)
    > Tag: SSID parameter set: "30 Munroe St"
    > Tag: Supported Rates 1, 2, 5.5, 11, 6, 9, 12, 18, [Mbit/sec]
    > Tag: Request
    > Tag: Extended Supported Rates 24, 36, 48, 54, [Mbit/sec]
    > Tag: Vendor Specific: Intel Corporation

```

如图，主机支持 1Mbps, 2Mbps, 5.5Mbps, 11Mbps, 6Mbps, 9Mbps, 12Mbps, 18Mbps, 24Mbps, 36Mbps, 48Mbps, 54Mbps

AP 支持 1Mbps, 2Mbps, 5.5Mbps, 11Mbps, 6Mbps, 9Mbps, 12Mbps, 18Mbps, 24Mbps, 36Mbps, 48Mbps, 54Mbps

16. What are the sender, receiver and BSS ID MAC addresses in these frames? What is the purpose of these two types of frames? (To answer this last question, you'll need to dig into the online references cited earlier in this lab).

```

IEEE 802.11 Association Request, Flags: .....C
Type/Subtype: Association Request (0x0000)
> Frame Control Field: 0x0000
  .000 0000 0010 1100 = Duration: 44 microseconds
> Receiver address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
> Destination address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
> Transmitter address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)
> Source address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)
> BSS Id: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  .... 0000 = Fragment number: 0
  0110 0111 0000 .... = Sequence number: 1648
  Frame check sequence: 0xfe3badc6 [unverified]
  [FCS Status: Unverified]
  [WLAN Flags: .....C]

```

如图，在请求帧中，

发送方 MAC 地址是 00:13:02:d1:b6:4f

接收方 MAC 地址是 00:16:b6:f7:1d:51

BSS ID 的 MAC 地址是 00:16:b6:f7:1d:51

```

IEEE 802.11 Association Response, Flags: .....C
Type/Subtype: Association Response (0x0001)
> Frame Control Field: 0x1000
  .000 0001 0011 1010 = Duration: 314 microseconds
> Receiver address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)
> Destination address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)
> Transmitter address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
> Source address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
> BSS Id: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  .... 0000 = Fragment number: 0
  1110 1001 0000 .... = Sequence number: 3728
  Frame check sequence: 0x37f2ab2b [unverified]
  [FCS Status: Unverified]
  [WLAN Flags: .....C]

```

如图，在响应帧中，

发送方 MAC 地址是 00:16:b6:f7:1d:51

接收方 MAC 地址是 00:13:02:d1:b6:4f

BSS ID 的 MAC 地址是 00:16:b6:f7:1d:51

问题及收获：

802.11 协议是一种无线局域网通信协议，它定义了无线网络的物理层和链路层的标准，使得不同厂商的设备能够进行互操作并实现无线通信。该协议支持多种频段和调制方式，提供了不同的传输速率和数据传输机制，如 CSMA/CA（载波监听多路访问/冲突避免）。802.11 协议的出现和发展，使得无线网络技术得到了广泛的普及，为人们提供了方便、灵活的网络接入方式。

建立在 802.11 协议上的 WiFi 技术，是目前最为普及的无线网络接入方式之一。WiFi 使得用户可以在无线网络覆盖范围内，通过笔记本电脑、智能手机、平板电脑等移动设备随时随地接入互联网，进行数据传输、浏览网页、观看视频、在线游戏等活动。它不仅满足了人们对网络接入的便利性需求，还推动了移动互联网时代的到来，改变了人们的生活方式和模式，促进了信息的快速流通和交流，对社会的信息化发展起到了重要的推动作用。